

Davi Guanabara Aragão

AI/ML Engineer | PhD Researcher in Autonomous Systems and Data Science

São Paulo, Brazil | davi_guanabara@live.com | +55 85 9

8673-6026 | github.com/DaviGuanabara | linkedin.com/in/davi-guanabara | lattes.cnpq.br/2180140366316472

PROFESSIONAL SUMMARY

AI/ML Engineer and PhD researcher at the Aeronautics Institute of Technology (ITA), applying artificial intelligence, machine learning, and software engineering to autonomous systems, geospatial computing, and aerospace applications. Currently contributes to research initiatives in collaboration with the Brazilian Airspace Control System (SAC/ICEA), developing computational methods for urban airspace planning, intelligent decision-making, and simulation-driven experimentation. Research interests focus on reinforcement learning, multi-agent systems, geospatial AI, and reproducible research software for complex engineering problems.

AREAS OF EXPERTISE

Autonomous Systems: Autonomous Decision-Making • Multi-Agent Systems • Autonomous UAV Systems

Machine Learning: Deep Learning • Reinforcement Learning • Attention-Based Models

Software Engineering: Modular Software Architecture • Event-Driven Systems • Experimentation Frameworks

Geospatial Computing: Geospatial Modeling • Spatio-Temporal Modeling

SELECTED PROJECTS

Deep Reinforcement Learning with Graph Attention for Cooperative Loyal Wingmen (2024--2026)

CTEDS Research Program | Brazil--Sweden Air Domain Study

- Developed a deep reinforcement learning framework for cooperative threat-engagement decision-making in Loyal Wingmen UAVs operating under shared spatio-temporal observations.
- Proposed a graph-attention neural architecture enabling adaptive information sharing and coordinated policy learning under partial observability.
- Designed and evaluated simulation-based experiments integrating graph neural networks with multi-agent reinforcement learning for autonomous aerial systems.
- Conducted within the Cooperative Threat Engagement with Heterogeneous Drone Swarms (CTEDS) research program, an international Brazil--Sweden collaboration involving government, academia, and industry.
- Resulted in two peer-reviewed publications:
Journal of Intelligent & Robotic Systems (Springer Nature, 2026, DOI:10.1007/s10846-026-02415-8) and *2024 Brazilian Symposium on Robotics (SBR) and Workshop on Robotics in Education*(DOI:10.1109/SBR/WRE63066.2024.10837749).

SkyWeaver --- Dynamic Airspace Configuration Platform for BRUTM (2025--Present)

Research Project in collaboration with SAC/ICEA

- Designed a reactive architecture supporting Dynamic Airspace Configuration to contribute with the Brazilian UTM (BRUTM) concept.
- Developed modular software components for geospatial scene generation, operational state management, routing, and airspace infrastructure modeling.
- Investigating graph-based optimization methods for droneport placement, route planning, and infrastructure provisioning under operational constraints.
- Ongoing research explores software engineering, optimization, and AI techniques for future UTM, U-Space, and Advanced Air Mobility (AAM) operations.

OndeVale --- A Property Valuation Platform (2025)

Cloud-Deployed AI System

- Developed a cloud-deployed platform for residential property valuation combining geospatial analytics, machine learning, and interactive web visualization.
- Designed the complete machine learning pipeline, including data preprocessing, feature engineering, model training, inference services, and deployment.
- Built a FastAPI backend and Vue.js frontend integrating GIS visualization with real-time model inference.
- Deployed the application as a publicly accessible cloud service, connecting data preprocessing, model inference, a REST API, and an interactive geospatial interface.

A Modular Framework for Reproducible Supervised Learning: A Vineyard Leaf Wetness Prediction Case Study (2024--2025)

- Developed a deep learning model for vineyard leaf wetness prediction using meteorological observations from the Iowa Environmental Mesonet.
- Designed a permutation-invariant Deep Sets architecture with masked attention to model heterogeneous spatio-temporal neighborhoods.
- Built a reproducible supervised-learning framework integrating stochastic contextual sampling, Bayesian hyperparameter optimization, bootstrap evaluation, and deterministic seed management.

EDUCATION

Ph.D. in Electronic Engineering and Computing (2025--Present)
Aeronautics Institute of Technology (ITA)

Graduate Certificate in Data Science (2024--2026)
Aeronautics Institute of Technology (ITA)

M.S. in Aeronautical Computing (2021--2024)
Aeronautics Institute of Technology(ITA)

B.S. in Control and Automation Engineering (2013--2019)
University of Fortaleza (UNIFOR)